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Effects of an external load on anticipatory mechanisms of handstand walking initiation in experienced gymnasts

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ABSTRACT

Anticipatory postural adjustments (APAs) are responsible for a successful first step execution in handstand walking. This study evaluates gymnasts' ability to adapt their APAs and stepping parameters in response to adding/removing an external load over repeated handstand walking initiation trials. Eighteen gymnasts performed five handstand walking initiation trials without load (PRE), eight trials with an external load (LOAD) and five trials with removed load (POST). Force platforms and a motion capture system were used to quantify mechanical parameters from the anticipatory phase and the stepping phase. During the first LOAD trial, APAs did not change, however, the step height decreased by 0.56 cm compared to PRE ($p = 0.007$). During the second and third LOAD trials, the mediolateral center of pressure APAs increased by 2.0 cm ($p = 0.01$) and 2.3 cm ($p = 0.01$) compared to the first LOAD trial, while the step parameters returned to baseline. The removed load did not change APAs, while the step parameters were initially altered and then returned to baseline. Results show gymnasts' ability to modify their APAs over repeated trials to adapt to an external load, enabling them to initiate accurate forward steps. Training exercises should target handstand walking preparatory mechanisms for a smooth transition between handstand and stepping.

HIGHLIGHTS

- Gymnasts demonstrate the ability to adapt their neuromechanical patterns of handstand walking initiation in response to external perturbations such as additional wrist loads.
- Ground reaction force, center of pressure and center of mass parameters can reliably assess a gymnast's postural adaptation to external perturbations. Adding external load to the stepping hand reduces the peak vertical displacement of the hand during step execution.
- Frontal plane parameters illustrate how gymnasts accommodate this new perturbation.

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Balance; centre of pressure; step; biomechanics; anticipatory postural adjustments; adaptation

Introduction

Handstand is an inverted vertical body position that is adopted across gymnastics disciplines (Fédération Internationale De Gymnastique, 2017) and cross-fit (Martínez-Gómez et al., 2020). The ability to take steps from handstand using the hands to displace the body, namely handstand walking, is an important skill progression to challenge gymnasts' balance ability (Fédération Internationale De Gymnastique, 2017) and as a strength exercise in the rehabilitation after arm injury (Pidcoe & Burnet, 2007).

The step characteristics of handstand walking exhibit a similar cadence to bipedal walking but shorter strides, a wider base of support and longer double support time (Earhart & Mosiman, 2010). The initial step transition from stable handstand balance mirrors a critical aspect of bipedal gait initiation, demanding a complex interplay of neural and biomechanical factors to maintain balance. Such complexity was addressed for the first time in a previous investigation on the anticipatory postural adjustments (APAs) preceding the first step execution of handstand walking (Grabowiecki et al., 2021). In particular,

a preparatory shift of the pressure applied to the ground through hand contact was shown to be essential for achieving the required whole-body centre of mass (CoM) velocity for hand-off. Nonetheless, the preparatory centre of pressure (CoP) shifted posteriorly and laterally toward the stepping limb to propel athletes' CoM anteriorly and toward the support limb, thus minimising balance disturbance during the initiation of the first forward handstand step.

While this behaviour was similar to that observed in bipedal gait initiation (for a review, see Winter, 1995; Yiou et al., 2017), the preparatory CoP displacement for handstand stepping displayed greater variability in the anteroposterior (AP) than in the mediolateral (ML) direction (Grabowiecki et al., 2021). These results are further supported by similar findings (Gautier et al., 2007; Hedbávný et al., 2013; Kerwin & Trewartha, 2001; Omorczyk et al., 2018; Puszczafowska-Lizis & Omorczyk, 2019; Slobounov & Newell, 1996; Sobera et al., 2007), who reported that COP displacement values in the AP direction during a handstand were greater than those in the ML direction. This variability is likely due to the mechanically less stable position inherent in handstand balance, characterised by a reduced

base of support, higher CoM and the reliance on the wrist joint strategy to maintain balance compared to the bipedal stance (Gautier et al., 2007; Hedbávný et al., 2013; Kerwin & Trewartha, 2001; Slobounov & Newell, 1996; Sobera et al., 2007). Therefore, targeted training exercises should encompass not only the execution of handstand walking but also focus on the preparatory mechanisms in the attempt to reduce variability and allow for a smooth transition between handstand and handstand stepping.

The characteristics of APAs are closely aligned with the specific properties of the voluntary movement that precedes them (Ahmed & Wolpert, 2009). These adjustments are modulated further in response to factors such as the uncertainty, magnitude and direction of a perturbation, as well as biomechanical constraints of the task (Ahmed & Wolpert, 2009; Caderby et al., 2017; Laudani et al., 2021; Mille et al., 2014; Monjo & Forestier, 2014; Patchay & Gahéry, 2003; Rum et al., 2021). It has been proposed that APAs rely on a feedforward mechanism, where an internal model representing body dynamics and environmental conditions is employed to predict movement outcome, thereby facilitating the appropriate postural adjustment in advance. Ahmed and Wolpert (2009) (Ahmed & Wolpert, 2009) further explored this concept in an arm-reaching task where participants held a resisted handle on a robotic arm. Their findings demonstrated that, with repeated exposure to a specific perturbation, individuals learned to anticipate the perturbation, adjusting both their arm movement control and postural responses accordingly.

Previous studies on bipedal gait initiation adopted unexpected, externally induced balance perturbations to investigate the person's ability to respond by correcting/amending their CoP preparatory displacement as well as the stepping characteristics (Laudani et al., 2021; Rum et al., 2021). Continual and repetitive application of external load-induced balance perturbations during movement disturbance can induce permanent adaptations in an individual's preparatory mechanism (Patchay & Gahéry, 2003). These findings suggest the presence of an adaptation which enables more effective responses to perturbations, thereby maintaining the intended movement execution.

According to the principles of training, exercises need to sufficiently challenge the relevant aspects of the intended task to prompt subsequent improvements in performance (Matveev, 1981). In addition, perturbation-based balance training during bipedal gait has demonstrated a notable reduction in fall incidents and enhanced balance (Mansfield et al., 2015). Introducing repeated balance perturbations in handstand walking could effectively *i)* investigate athletes' ability to respond to the perturbation and adapt over trials, *ii)* enhance individual's capability to manage external constraints affecting the task. This approach holds potential in helping coaches and trainers to design tailored perturbation-based training exercises aimed at enhancing balance and performance and facilitating athletes' adaptability to external constraints during handstand walking initiation. In this study, we addressed this issue by applying an external load to the forearm of the stepping limb during handstand walking initiation, maintaining the same load across repeated trials.

This study aimed to evaluate the effects of adding/removing an external load applied to the wrist on preparatory and executive postural adjustments of handstand walking initiation. Based on our previous study's finding showing the crucial role of anticipatory mechanisms for a successful handstand step initiation (Grabowiecki et al., 2021), it was hypothesised that repeatedly adding an external load would cause an initial disruption of the first step execution followed by a trial-by-trial adaptation of the anticipatory postural adjustments allowing for a return to the baseline step characteristics. Based on basic motor learning principles and previous studies' results on repeated loads (Ahmed & Wolpert, 2009; Monjo & Forestier, 2014), it was also expected that removing the load would initially cause a disturbance, followed by a gradual return to baseline level of the step characteristics.

Methods

Participants

A convenience sample of 18 healthy gymnasts (age: 23.2 ± 4.6 years; body mass: 61.2 ± 9.6 kg; height: 1.69 ± 0.06 m; gymnastic experience: 13.9 ± 5.4 years) volunteered to participate in this study. The sample size was based on an a priori power analysis using the G*Power software. The analysis employed repeated measures analysis of variance (RM ANOVA) with within-subject factors as the selected statistical test used. The analysis was based on an α error probability of 0.05, a statistical power ($1-\beta$) of 0.80 and an effect size f of 0.29. Inclusion criteria comprised a minimum of 5 years of gymnastic experience, ability to maintain a stable handstand position for 10 s and the ability to execute four controlled consecutive handstand steps. Participants were also required to have no history of upper body injuries. Both left- and right-handed gymnasts were included in the study. Written informed consent was obtained before inclusion in the study. The procedures were approved by the institutional review board of the University of Rome "Foro Italico" (approval code CAR18/2020). All testing was conducted in a university laboratory setting.

Protocol

Warm-up included three 5-second handstand holds, three sets of four handstand steps and supplementary stretching of the upper and lower limbs. Following the warm-up, participants performed handstand walking initiation trials under three experimental conditions adhering to the following sequence: *i)* five trials without external load (PRE); *ii)* eight trials with an external load placed on the dominant limb forearm (LOAD); *iii)* five trials following the removal of the load (POST). During all conditions, participants were instructed to hold a stable handstand position for 3 s before initiating a handstand step with their dominant hand at their most comfortable speed (Caderby et al., 2017). The dominant hand was identified by each participant according to their preferred hand for initiating a handstand walking step. The experimenter ensured that a handstand step was correctly performed with the participant picking up the movement of one hand and placing it back down

onto the ground, performing at least four consecutive handstand steps forward. The starting position of the hands was marked on the ground, and participants were instructed to maintain the same initial hand position for all trials. During the LOAD condition, participants wore an armband at the middle of their dominant forearm length; the armband was filled with an external load of 3% participant's body mass. This threshold was set by the researchers after prior testing aimed at identifying the external load needed to elicit notable changes in the CoP during handstand walking initiation without generating discomfort to the participant. Blood flow and range of movement at the wrist and elbow joints were not restricted. A three-minute rest was given between each trial. Gymnasts were asked to rest sitting on a chair and placing the stepping limb on a table without moving it. A trial was accepted for analysis when the participant maintained a stable handstand position for 3 s with the body held vertically in line and with legs together, and four forward steps were correctly performed (Fédération Internationale De Gymnastique, 2017).

Instrumentation

Two piezoresistive force plates (40 × 40 cm Bertec Corp., Columbus, Ohio, USA) were positioned adjacent to each other at ground level and gymnasts placed one hand on each force plate. Ground reaction forces (GRF) and CoP data were collected at a sampling rate of 1000 hz and recorded using motion analysis software (VICON Nexus 2.1, Oxford Metrics London, UK). The net CoP from both force plates was computed (Winter et al., 1996). A seven-camera stereophotogrammetric system (MX3 System, Vicon Motion Systems, Los Angeles, USA) collected the position of 39 retro-reflective spherical markers located on the participants' skin to create a 15-segment bio-mechanical model (Plug-in-Gait full-body model) (Grabowiecki et al., 2021; Kadaba et al., 1990) with a 100 hz sampling rate. Height and weight were collected using a stadiometer and a weight scale (Seca 700, Seca, Germany).

Data processing

Force plates and motion capture data were filtered using a second-order low-pass Butterworth filter with a cut-off frequency of 15 hz (Grabowiecki et al., 2021; Rum et al., 2021). All data were analysed using a custom Matlab code (MATLAB R2020a; MathWorks, Natick, MA) to define handstand walking initiation phases and extract parameters of interest (Table 1). Each trial was divided into a preparation phase (from APA onset to hand-off) and an execution phase (from hand-off to hand-contact). Parameters of interest were obtained from the postural and the executory phases as outlined in Table 1. The base of support (BoS) was defined by the area covered by the hands on the ground and estimated using finger and wrist marker coordinates of each hand.

Statistical analysis

Intra-subject coefficients of multiple correlations of AP and ML CoP displacement were used to select three trials from the PRE condition with the highest similarity for each participant (Grabowiecki et al., 2021). The parameters extracted from these three trials were averaged to obtain a PRE block as representative of the preload condition and used for further statistics. The last three trials of the LOAD condition were averaged to create a lateLOAD block. The first three trials of the POST condition were averaged to create an earlyPOST block.

Data distribution was assessed using the Shapiro – Wilk test. Repeated measures ANOVA followed by post-hoc analysis with Holm Bonferroni-correction (Holm, 1979) for pairwise comparisons were used to evaluate the difference between conditions and trials for dependant variables (Table 1). Sphericity assumption was checked by Mauchly's test and, when violated, the Greenhouse – Geisser adjustment was applied on repeated measures. Nonparametric Friedman ANOVA with post hoc two-tailed Wilcoxon signed-rank test was used for not normally distributed parameters.

Table 1. Definition of handstand walking initiation phases, ground reaction force (GRF), centre of pressure (CoP), centre of mass (CoM) and step characteristics parameters.

Phases	APA Onset	The time instant when the ML CoP velocity exceeded 3 standard deviations above the mean calculated over a 0.5-second baseline period Chen, et al. (2019)
	Hand-off	The time instant when the vertical GRF beneath the stepping limb reached 0 N
	Hand-contact	The lowest vertical position of the marker placed on the middle knuckle of the stepping limb following the hand-off
GRF	ML-GRF	The peak lateral GRF toward the stepping limb during the Preparation phase (normalised to body weight)
	V-GRF	The peak Vertical GRF beneath the stepping limb during the Preparation phase (normalised to body weight)
CoP	CoP position at APA onset	The CoP at APA onset in mediolateral and anteroposterior direction, expressed as a percentage of the BoS boundaries
	AP CoP Displacement	The maximal posterior and maximal anterior CoP displacement during the Preparation phase
	ML CoP Displacement	The maximal mediolateral CoP displacement towards the stepping limb during the Preparation phase
	APA Duration	The time from handstand walking initiation onset until maximal mediolateral CoP displacement
CoM	CoM position at APA onset	The CoM at APA onset in the mediolateral and anteroposterior direction, expressed as a percentage of the BoS boundaries
	AP/ML CoM at hand-off and hand-contact	The anteroposterior and mediolateral CoM displacement at hand-off and hand-contact, calculated with respect to CoM position at APA onset
Step Characteristics	Step Height	The difference between the vertical position of the stepping limb marker at APA onset and its peak vertical position during step execution.
	Step Length	The step length and width were defined as anteroposterior and mediolateral distance between finger markers of right- and left-hand during Hand-contact, respectively
	Step Time	The duration from hand-off to hand-contact
	Step Velocity	The average step velocity, calculated by dividing step length by step time

To address the study's specific aims, appropriate trials were selected and organised into specific blocks for distinct comparisons. Specifically, to evaluate the effects of adding the external load during handstand walking initiation, the PRE block was compared to each of the first three trials of the LOAD condition and with the mean of the lateLOAD block (i.e., when the effect of the load was acquired and stable). To evaluate the early adaptation to the external load, the first trial in the LOAD condition (L1) was compared with the second and third LOAD trials (L2 and L3, respectively). To evaluate the effect of the removal of the load, the first three POST trials and the earlyPOST block were compared to the lateLOAD block. To evaluate the return to baseline following the load removal, the earlyPOST block and the last two trials of the POST condition were compared with the PRE block. Partial eta squared (η^2_p) and Kendall's W (W) were respectively used to report effect size for repeated measures ANOVA and nonparametric Friedman ANOVA. The significance level α was set to 0.05. All analyses were performed using the SPSS 26.0 software (Chicago, IL, USA).

Results

Base of support

There was a main effect of condition between PRE and LOAD in ML CoP position relative to BoS ($F_{1,428} = 4.25$, $p = 0.037$, $\eta^2_p = 0.2$) (Figure 1). ML CoP position relative to BoS significantly increased by 4.7% towards the stepping limb during

lateLOAD compared to PRE ($p = 0.004$). No main effect of condition was found between PRE, LOAD and POST in other BoS-related variables (Figure 1).

Ground reaction forces

There was a main effect of condition between PRE and LOAD in ML-GRF ($\chi^2(3) = 10.876$, $p = 0.012$, $W = 0.201$) and V-GRF ($F_{3,51} = 12.951$, $p < 0.001$, $\eta^2_p = 0.432$) (Figure 2). ML-GRF increased by 22.0% during L2 compared to PRE ($p = 0.002$). V-GRF increased by 11.3% during L2 ($p < 0.001$) and 11.2% during L3 ($p < 0.001$) compared to PRE. V-GRF also increased by 7.3% during L2 ($p = 0.014$) and 7.2% during L3 ($p = 0.009$) compared to L1. The removal of the load did not affect the ML-GRF. However, there was a main effect of condition in V-GRF between PRE, lateLOAD and earlyPOST ($F_{2,34} = 10.281$, $p < 0.001$, $\eta^2_p = 0.377$). V-GRF increased by 8.1% during lateLOAD ($p = 0.001$) and 3.9% during earlyPOST ($p = 0.014$) compared to PRE. V-GRF decreased by 4.2% during earlyPOST compared to lateLOAD ($p = 0.048$). There was no main effect in any of the GRF parameters between PRE and the last two trials of POST.

CoP displacement and APA duration

There was a main effect of condition between PRE and LOAD for ML CoP displacement ($\chi^2(3) = 8.333$, $p = 0.04$, $W = 0.154$) (Figure 3). No significant difference was found between PRE and the first three LOAD trials, whereas ML CoP displacement

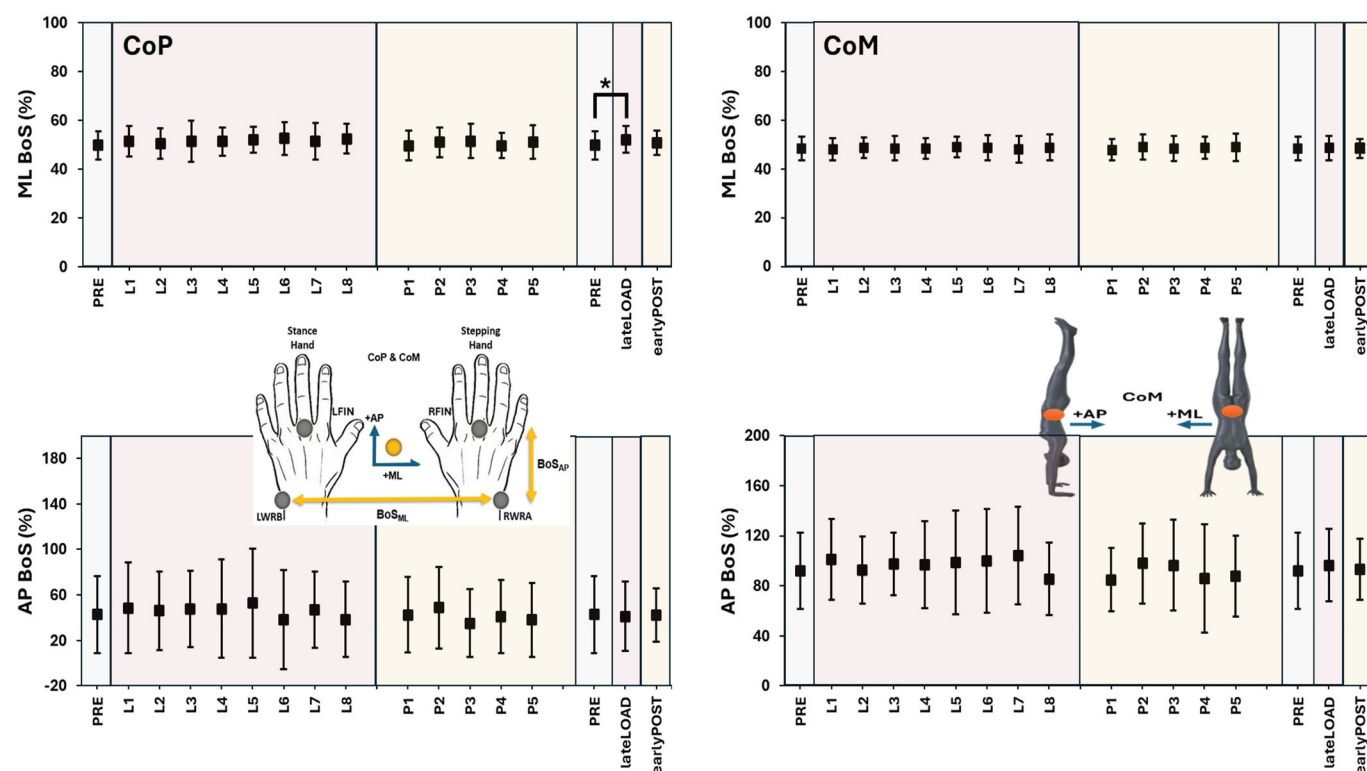


Figure 1. Group averages ± 1 SD of ML CoP, ML CoM, AP CoP and AP CoM initial position at APA onset presented as percentage of BoS. PRE represents the average of three trials at baseline without the load. L1-L8 are the eight trials of the LOAD condition. P1-P5 are the five trials of the POST condition. The last three trials of the LOAD condition were averaged to create a lateLOAD block. The first three POST condition trials were averaged and presented as earlyPOST. A positive value for ML CoP/CoM BoS corresponds to the initial position towards the stepping limb direction. A positive value for AP CoP/CoM BoS corresponds to the initial position towards anterior direction. *Significant difference between trials or block ($p < 0.05$). **Significant difference between trials or block ($p < 0.001$).

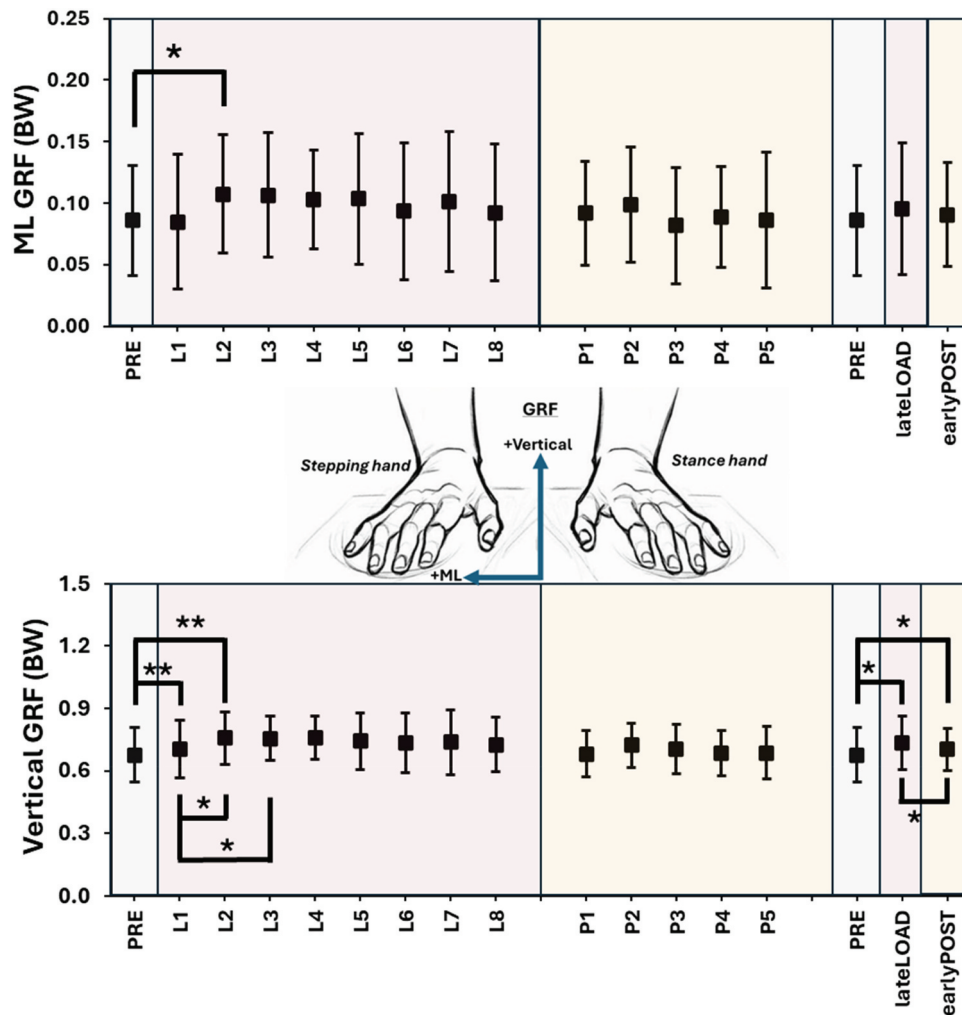


Figure 2. Group averages ± 1 SD of Normalised Mediolateral GRF (ML GRF) and Normalised Vertical GRF (VGRF). For figure legend please refer to [Figure 1](#) caption. A positive value for ML GRF corresponds to the movement towards the stepping limb direction.* Significant difference between trials or block ($p < 0.05$). **Significant difference between trials or block ($p < 0.001$).

towards the stepping limb increased by 28.7% during L2 ($p = 0.01$) and 32.1% during L3 ($p = 0.006$) compared to L1. There was no main effect of condition between PRE, LOAD and POST in anterior and posterior CoP displacements and APA duration.

CoM displacement

There was a main effect of condition between PRE and LOAD for ML CoM displacement at hand-off ($F_{3,51} = 7.013$, $p < 0.001$, $\eta^2_p = 0.292$) and ML CoM displacement at hand-contact ($F_{3,51} = 7.332$, $p < 0.001$, $\eta^2_p = 0.301$) ([Figure 4](#)). ML CoM displacement towards the stance limb at hand-off increased by 24.7% during L2 ($p = 0.002$) and 22.1% during L3 ($p = 0.009$) compared to PRE; the same parameter also increased by 27.6% during L2 ($p < 0.001$) and 25.0% during L3 ($p = 0.006$) compared to L1. ML CoM displacement towards the stance limb at hand-contact increased by 60.5% during L2 ($p = 0.013$) and 62.0% during L3 ($p = 0.011$) compared to PRE; the same parameter also increased by 89.0% during L2 ($p < 0.001$) and 90.3% during L3 ($p = 0.003$) compared to L1. There was no main effect of condition between PRE, LOAD and POST in AP CoM displacement

parameters. There was no main effect in any of the CoM parameters between PRE and the last two POST trials.

Step characteristics

There was a main effect of condition between PRE and LOAD in step height ($F_{3,51} = 3.586$, $p < 0.02$, $\eta^2_p = 0.174$) ([Table 2](#)). Step height decreased by 8.9% during L1 ($p = 0.007$) compared to the PRE. For the same parameter, there was a main effect of condition between PRE, lateLOAD and earlyPOST ($F_{2,34} = 3.498$, $p = 0.042$, $\eta^2_p = 0.171$). Step height decreased by 5.8% during the lateLOAD compared to the PRE ($p = 0.014$). Similarly, the step height increased by 5.9% during the earlyPOST compared to the lateLOAD ($p = 0.016$). However, no significant difference was found between PRE and the earlyPOST. No main effect of condition (PRE, LOAD and POST) was observed in step length. There was a main effect of condition between PRE and LOAD in step width ($F_{3,51} = 3.586$, $p = 0.02$, $\eta^2_p = 0.217$). No significant difference was found between PRE and the first three LOAD trials, whereas the step width significantly decreased by 7.4% during L2 compared to L1 ($p = 0.008$). For the same parameter, there was a main effect of condition between PRE, lateLOAD

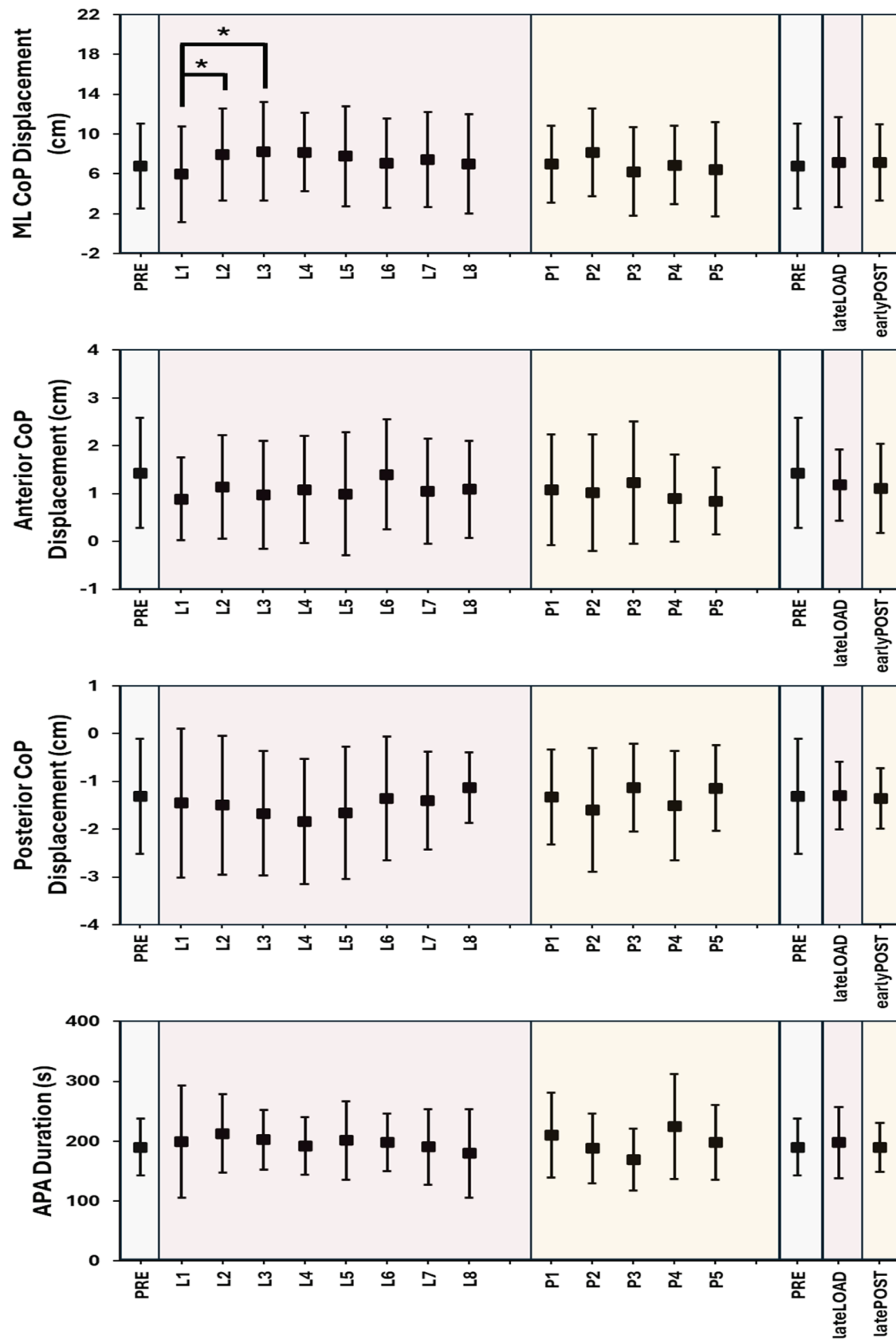


Figure 3. Group averages ± 1 SD of ML CoP Displacement, Anterior CoP Displacement, Posterior CoP Displacement and APA duration. For figure legend please refer to Figure 1 caption. A positive value for ML CoP corresponds to the movement towards the stepping limb direction. *Significant difference between trials or block ($p < 0.05$). **Significant difference between trials or block ($p < 0.001$).

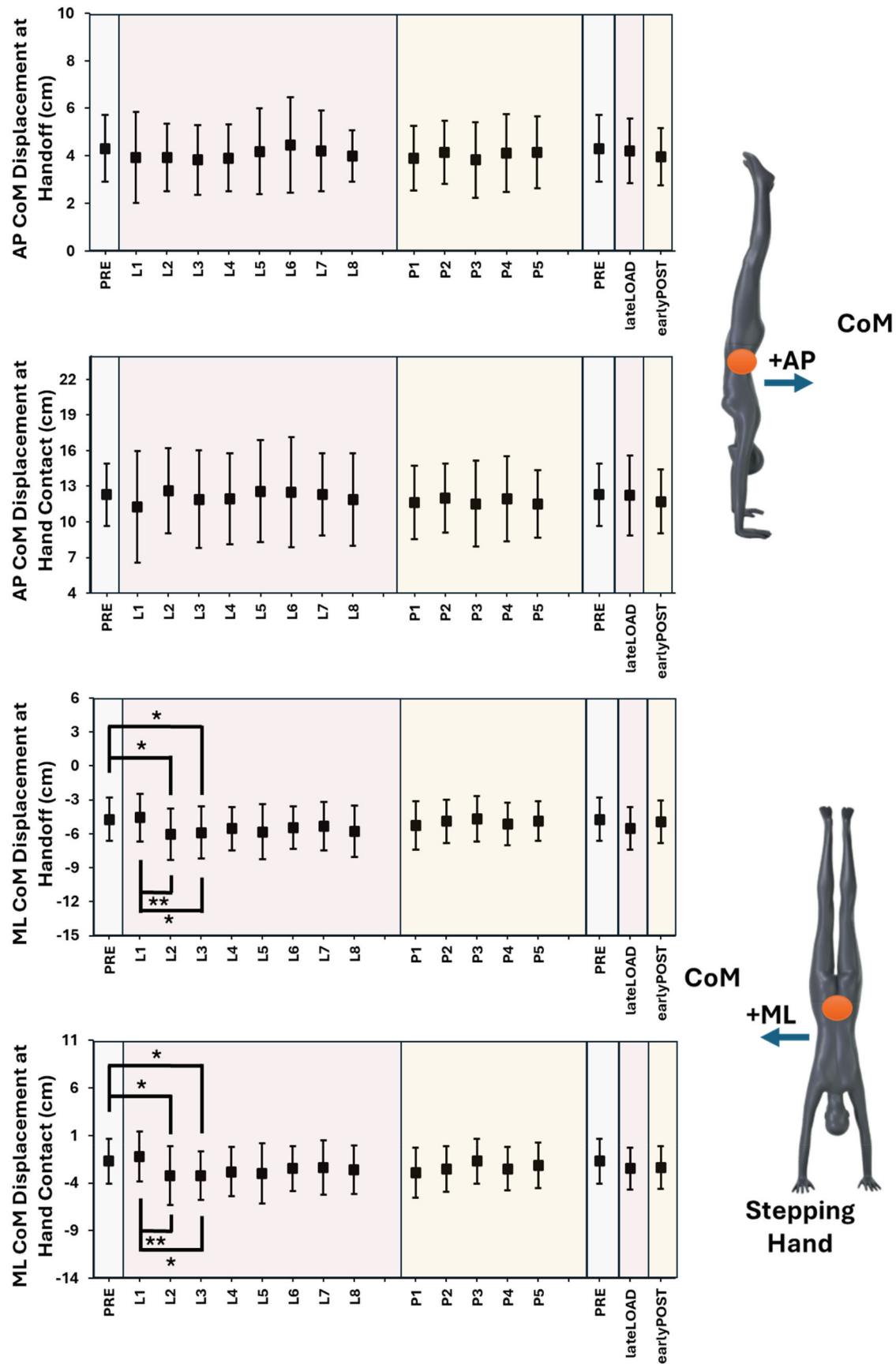


Figure 4. Group averages ± 1 SD of AP CoM Displacement at Hand-off, AP CoM Displacement at Hand-Contact, ML CoM Displacement at Hand-off, ML CoM Displacement at Hand-Contact. For figure legend please refer to Figure 1 caption. A positive value for AP CoM corresponds to the movement towards anterior direction. A positive value for ML CoM corresponds to the movement towards the stepping limb direction. *Significant difference between trials or block ($p < 0.05$). **Significant difference between trials or block ($p < 0.001$).

Table 2. Group averages ± 1 SD of Step Height, Step Length, Step Width, Step Time and Step Velocity. For table legend please refer to Figure 1 caption. *Significant difference between Pre and L1 conditions ($p < 0.05$). †Significant difference between Pre and lateLOAD block ($p < 0.05$). ‡Significant difference between lateLOAD and earlyPOST block ($p < 0.05$). §Significant difference between L1 and L2 conditions ($p < 0.05$).

	PRE	L1	L2	L3	L4	L5	L6	L7	L8	P1	P2	P3	P4	P5	lateLOAD	earlyPOST
Step Height (cm)	6.57 $\pm$ 1.55^{††}	6.01 $\pm$ 1.42[*]	6.86 $\pm$ 1.70	6.34 $\pm$ 1.82	6.24 $\pm$ 1.71	6.48 $\pm$ 1.60	6.18 $\pm$ 1.69	32.35 $\pm$ 6.83	6.04 $\pm$ 1.22	6.75 $\pm$ 1.14	6.56 $\pm$ 1.16	6.41 $\pm$ 1.33	6.5 $\pm$ 1.31	6.46 $\pm$ 1.68	6.20 $\pm$ 1.28^{††}	6.57 $\pm$ 1.07[*]
Step Length (cm)	31.38 $\pm$ 5.71	30.83 $\pm$ 8.56	34.29 $\pm$ 7.05	31.68 $\pm$ 8.82	32.36 $\pm$ 8.16	33.09 $\pm$ 8.38	31.75 $\pm$ 7.55	32.35 $\pm$ 6.83	31.87 $\pm$ 7.14	30.06 $\pm$ 7.69	31.27 $\pm$ 5.97	30.58 $\pm$ 7.83	31.64 $\pm$ 7.03	31.29 $\pm$ 6.16	31.99 $\pm$ 6.24	30.64 $\pm$ 6.67
Step Width (cm)	46.92 $\pm$ 6.00	48.54 $\pm$ 4.04[†]	45.08 $\pm$ 7.20[†]	45.91 $\pm$ 7.60	46.50 $\pm$ 6.20	46.32 $\pm$ 5.89	47.35 $\pm$ 7.47	47.29 $\pm$ 6.03	46.24 $\pm$ 6.04	45.34 $\pm$ 7.43	45.59 $\pm$ 5.95	46.20 $\pm$ 6.89	45.05 $\pm$ 6.90	46.02 $\pm$ 8.03	46.96 $\pm$ 6.20	45.71 $\pm$ 6.42
Step Time (s)	0.27 $\pm$ 0.04	0.26 $\pm$ 0.03	0.30 $\pm$ 0.06	0.29 $\pm$ 0.06	0.28 $\pm$ 0.04	0.28 $\pm$ 0.05	0.27 $\pm$ 0.03	0.28 $\pm$ 0.04	0.28 $\pm$ 0.05	0.28 $\pm$ 0.04	0.26 $\pm$ 0.05	0.27 $\pm$ 0.06	0.27 $\pm$ 0.05	0.26 $\pm$ 0.04	0.28 $\pm$ 0.04	0.27 $\pm$ 0.04
Step Velocity (m/s)	1.19 $\pm$ 0.35	1.19 $\pm$ 0.32	1.17 $\pm$ 0.31	1.14 $\pm$ 0.37	1.18 $\pm$ 0.31	1.19 $\pm$ 0.31	1.19 $\pm$ 0.34	1.20 $\pm$ 0.39	1.16 $\pm$ 0.27	1.12 $\pm$ 0.34	1.23 $\pm$ 0.33	1.20 $\pm$ 0.43	1.20 $\pm$ 0.36	1.24 $\pm$ 0.34	1.18 $\pm$ 0.30	1.19 $\pm$ 0.35

and earlyPOST ($F_{2,34} = 4.524$, $p = 0.018$, $\eta^2_p = 0.210$). No significant difference was found between PRE and the lateLOAD as well as earlyPOST. Step width decreased by 2.7% during the earlyPOST compared to the lateLOAD ($p = 0.005$). There was no main effect in step width between PRE and the last two POST trials. There was a main effect of condition between PRE and lateLOAD in step time ($F_{3,51} = 4.000$, $p = 0.012$, $\eta^2_p = 0.190$). Step time increased by 15.0% during L2 compared to L1 ($p = 0.002$). There was no main effect of condition between PRE, LOAD and POST in Step Velocity.

Discussion

In the present study, for the first time to the authors' knowledge, gymnasts' ability to initiate handstand walking when an external load is applied and then removed to the forearm of the stepping limb over repeated trials was investigated. In accordance with our first hypothesis, step parameters were altered when an external load was added to the stepping limb during handstand walking initiation. Gymnasts demonstrated an early trial-by-trial adaptation of the postural adjustments to support the return to baseline step characteristics. In line with our second hypothesis, removing the load disrupted the first step characteristics during the early trials due to the newly acquired postural adaptation in the preparatory phase of the task and was followed by a gradual return to baseline for preparatory postural mechanisms and step characteristics. These results highlighted the importance of the preparatory mechanisms to allow for a successful handstand step initiation to adapt for changes in load (hence inertial) conditions throughout the movement. Therefore, it is speculated that handstand walking exercise protocols should consider not only the execution of movement itself, i.e., the action of initiating the stepping movement forward, but also the anticipatory phase that precedes and prepares the forthcoming movement execution.

Early adaptation to the external load

The external load attached to the middle of the forearm of the stepping limb alters the limb's inertial properties, compelling the central nervous system to recalibrate the internal representation of this body segment and its movements concerning environmental conditions.

This recalibration is vital for APAs required for the successful execution of initiating a forward handstand step. During the first LOAD trial, gymnasts could not predict how the external load changed their internal representation of the stepping limb, and therefore, their central nervous system was unable to anticipate the movement perturbation and activate appropriate APAs. As a result, a significantly lower step height was observed. Consistent with our hypotheses, subsequent trials revealed adaptive changes in postural control, reflecting alterations in APAs to accommodate the constraints imposed by the loaded condition. The adaptive feedback information obtained from the first loaded trial was used to recalibrate the motor programs with feedforward modifications that were associated with the intended movement. During the second and third LOAD trials, gymnasts increased their vertical GRF under the

stepping limb to overcome the additional load. This can be viewed as a strategy aimed at directly supporting the lifting of the stepping limb. Increased vertical and mediolateral GRF under the stepping limb resulted in a greater lateral shift of the CoP towards the stepping hand before the hand-off. These adjustments align with previous studies exploring body weight distribution in bipedal gait initiation (Caderby et al., 2017; Patchay & Gahéry, 2003). Adjustments in mediolateral GRF and CoP shift were shown to be fundamental to finely tune APAs in the event of unexpected perturbations and allow for successful step execution during gait initiation performed by young adults (Lyon & Day, 1997; Mille et al., 2014). From a biomechanical perspective, an external load at the stepping side will generate a greater lateral fall of the body towards the same side during the step execution. In this study, an increase in mediolateral CoM displacement towards the stance limb at hand-off and hand-contact during the second and third LOAD trials might indicate a postural strategy aiming to prevent an excessive lateral fall of CoM during the step execution (Bastian, 2008; Grabowiecki et al., 2021). This strategic adjustment appears to enable gymnasts to achieve a similar step height to the baseline, facilitated by an increased mediolateral CoP shift towards the stepping limb, in line with the principles of the inverted pendulum model for posture maintenance (Grabowiecki et al., 2021).

Interestingly, gymnasts exhibited consistent mechanical behaviour in the sagittal plane despite the introduction of the external load. Large between-subject variability of the antero-posterior CoP and CoM parameters could explain the lack of significant findings in the sagittal plane. This agrees with a recent study (Grabowiecki et al., 2021) reporting greater movement variability in the sagittal plane compared to the frontal plane during handstand walking initiation. Similar findings were also reported during bipedal gait initiation, where young healthy adults did not alter their sagittal plane behaviour when faced with an external medio-lateral disturbance that shifted their body-weight towards the stepping side (Caderby et al., 2017; Patchay & Gahéry, 2003). The authors also reported that the CoM initial position was significantly shifted towards the externally loaded side. On the contrary, the present results showed that the initial position of CoM with respect to BoS did not differ between conditions. However, in the current study, the load was lighter (3% BW vs 10% BW) and placed at a lower distance (forearm vs waist) with respect to the hinge. Future investigations should explore how varying loads and application positions influence adaptive postural strategies during handstand walking initiation.

Late adaptation to the external load

During the lateLOAD block, gymnasts still initiated the movement with a lower step height but with similar mediolateral APAs (GRF, CoP/CoM displacement) compared to PRE. In addition, in the lateLOAD block, the initial CoP position was closer to the BoS boundary of the stepping limb side compared to baseline, with a greater vertical GRF being exerted under the same side. Altogether, these results may indicate two aspects of the late postural adaptation to the external load. First, the mediolateral APAs seem to be responsible for adjusting the

step height during the externally induced perturbation. Without altering the mediolateral APAs (i.e., increasing mediolateral GRF and CoP towards the stepping limb as well as CoM shift towards the stance limb) gymnasts cannot overcome the perturbation effect of the external load and, therefore, a lower step is performed. Second, the change in the initial CoP position may be a result of the repeated exposure to the external perturbation which suggests a learning and adaptation process to the postural effects of the external load. This would indicate not only a short-term adaptation but also a completely new strategy adopted by the gymnasts to cope with the balance constraints set by the external load. Supporting this interpretation, previous literature already reported that humans are capable of adapting to a variety of perturbations when repeated exposure occurs (Ahmed & Wolpert, 2009; Martin et al., 1996; Mille et al., 2014; Monjo & Forestier, 2014).

Effects of removing the external load

The higher and narrower first step and decreased vertical GRF under the stepping limb during earlyPOST block compared to lateLOAD block suggests that step characteristics are altered as a result of the external load being removed from the stepping limb. However, the lack of changes in mediolateral APAs indicates that gymnasts were unable to instantly re-modulate their lateral postural adjustments in response to load removal. When systematic lateral pull perturbations were applied and then suddenly removed during bipedal gait initiation, different adaptive postural strategies such as an increase in mediolateral CoM displacement and wider step occurred mainly in the first trial and were immediately readapted to pre-perturbation levels (Mille et al., 2014). Unexpectedly, in the current study, no significant changes occurred in the analysed parameters at the first three POST trials when compared individually to lateLOAD. One reason could be that, in the study by (Mille et al., 2014), the disruptive mechanical effect of the perturbation was greater due to the different location of application (i.e., at the pelvis) and load magnitude (~10% of body weight) compared to the present study. These differences could have hence led to a longer-lasting adaptation of postural adjustments even after the perturbation load was removed. Another potential reason for the lack of mediolateral APAs changes in the POST condition trials could be the occurrence of fatigue (i.e., gymnasts performing the movement pattern with minimal energy demand, unable to modify their mediolateral APAs promptly and adapting to the absence of the LOAD).

Practical applications

The implications of our study findings hold significant value, particularly in the context of training handstand walking skills in sport disciplines like gymnastics and CrossFit. The results from our recent study (Grabowiecki et al., 2021) have highlighted that APAs are fundamental to preparing and executing handstand walking initiation as accurately as possible. With the present study results, we expand and build upon this concept by demonstrating that the adaptability of APAs preceding handstand stepping is a key factor allowing for appropriate stepping initiation under unpredictable environmental disturbances.

Noteworthy, in the current study, adding and removing the load did not result in performance failure. Hence, it is reasonable to assert that incorporating load attachment and removal during handstand walking initiation serves as a safe and effective training method for coaches and athletes seeking to diversify training modalities. This information can be used in structuring normal and perturbation-based balance training for handstand walking, and improving individual performance, balancing ability, and decreasing the chance of falls. The novelty of this study and the findings can provide the basis for further neuromechanical and motor control research on the topic.

Limitations and conclusions

A potential limitation of this study was the possibility of inducing muscle fatigue in participants throughout the repeated handstand walking initiation trials. In spite of the application of resting periods to reduce fatigue, it is still possible to argue that the experimental protocol induced a certain level of muscle fatigue in some of the participants. It should be highlighted that all gymnasts recruited in this study were experienced athletes performing regular exercise sessions on a weekly basis, which should indicate an adequate level of fitness and endurance. With this regard, future studies might specifically address this issue by including a varied (e.g., lower and greater) number of trials in their protocol, aiming to investigate how fatigue affects the ability to modify APAs and the step characteristics during handstand walking initiation. While gait speed may have influenced the results and potentially limited our findings, the step velocity analysis revealed no significant differences across trials, suggesting that step velocity was executed consistently and repeatably. Nevertheless, the lack of strict control over the initial handstand walking step speed remains a potential limitation. Given that gait speed is known to affect APAs and stepping parameters during walking, even subtle variations could have influenced the outcomes. Future studies should examine how different handstand walking step speeds impact APAs and step characteristics during the initiation phase of handstand walking.

In conclusion, the results of the present study show that gymnasts are capable of developing adaptive postural strategies when an external load is added during handstand walking initiation. In particular, gymnasts can learn by effectively accounting for the inertial properties of the external load and alter their APAs accordingly to restore adequate first step characteristics over trials.

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